

Example-2:

$$R = \{A, B, C, D, E\} \quad F = (A \rightarrow C, C \rightarrow BD, D \rightarrow A)$$

Let's find out α, β from the given

$$\alpha = \{E\}$$

$$\beta = \{B\}$$

	A	B	C	D	E
R $\rightarrow$	A	B	C	D	E
L $\rightarrow$	A	D	C	D	E

Iteration-1

$\alpha^+ = (E)^+ = E$ which is not super key

hence not candidate key

Iteration-2 (first iteration fails, leave)

So, now, we have to find out attribute closure by

taking single element from $(R - \beta) = (A, C, D, E)$ with E

$$(EA)^+ = EACBD ; \text{ so its SK, hence CK}$$

$$(EC)^+ = ECBD A ; \text{ so its SK, hence CK}$$

$$(ED)^+ = EDACB ; \text{ so its SK, hence CK}$$

$$\therefore CK = \{EA, EC, ED\}$$

Practice-2

$$R = (A, B, C, D, E)$$

$$F = (A \rightarrow C, B \rightarrow D, AC \rightarrow D, CD \rightarrow E, E \rightarrow A)$$

$$\alpha = \{B\}$$

$$A = \{ \}$$

Iteration-1

$$\alpha^+ = (B^+)^+ = BD ; \text{ not SK, not CK.}$$

Iteration-2 :

$$(BA)^+ = BACDE ; \text{ it is SK, hence CK.}$$

$$(BC)^+ = BCDEA ;$$

$$(BD)^+ = BD ; \text{ not SK, hence not CK}$$

$$(BE)^+ = BEDAC ; \text{ it is SK, hence CK}$$

Iteration-3: jisse janno pai nai, taake niye karo

$$(BDA)^+ = BDACE ; \text{ it is SK, hence CK}$$

$$\cancel{BDB}$$

$$(BDC)^+ = BDCEA ; \text{ it is SK, hence CK}$$

$$\cancel{BDD}$$

$$(BDE)^+ = BDEAC ; \text{ it is SK, hence CK}$$

$$CK = \{BA, BC, BE, BDA, BDC, BDE\}$$

* If there is no element in d , after first iteration, in 2nd iteration take one element at a time.

Canonical form / Minimal form :

Example -

$$R = \{A, B, C, D, E, F\}$$

$$F = \{ABC \rightarrow CDEF, C \rightarrow E, A \rightarrow B, D \rightarrow F\}$$

Step-1: (Decomposition state) (regular right side see more than one, shagula bhengee likho)

$$ABC \rightarrow C$$

$$ABC \rightarrow D$$

$$ABC \rightarrow E$$

$$ABC \rightarrow F$$

$$C \rightarrow E$$

$$A \rightarrow B$$

$$D \rightarrow F$$

Step-2: (Trivial remove)

$$X \ ABC \rightarrow C$$

$$ABC \rightarrow D$$

$$ABC \rightarrow E$$

$$ABC \rightarrow F$$

$$C \rightarrow E$$

$$A \rightarrow B$$

$$D \rightarrow F$$

nije nijeta generate korb, baad.

$$S = \{S\} : S \leftarrow S$$

$$A = \{A\} : A \leftarrow A$$

$$C = \{C\} : C \leftarrow C$$

$$ABC \rightarrow A = \{ABC\} : ABC \leftarrow A$$

$$ABC \rightarrow A = \{ABC\} : ABC \leftarrow A$$

$A \rightarrow D$

$A \rightarrow B$

$C \rightarrow E$

Attribute

Step-3: (Extraneous Remove)

$ABC \rightarrow D$

$ABC \rightarrow E$

$ABC \rightarrow F$

$C \rightarrow E$

$A \rightarrow B$

$D \rightarrow F$

helpers

Formula:

i) $A \rightarrow BC$

$B \rightarrow C$

ii) $*AB \rightarrow C, B \rightarrow C$

iii) $AB \rightarrow C, A \rightarrow B$

Final answer:

$C \rightarrow E$

$A \rightarrow B$

$D \rightarrow F$

$C \rightarrow E$ (repeated)

$AC \rightarrow D$

$AC \rightarrow F$

Let e ta jodi more chara, no take.
Dekhbo konoto extraneous kina.

$(C)^+ = CE$

$(A)^+ = AB$

$AC \rightarrow F$

$(C)^+ = CE$

$(A)^+ = AB$

Step-4: (Remove Functional Dependency extraneous)

$C \rightarrow E; (C)^+ = C$

$A \rightarrow B; (A)^+ = A$

$D \rightarrow F; (D)^+ = D$

$AC \rightarrow D; (AC)^+ = ACBEF$

$AC \rightarrow F; (AC)^+ = ACEBDF$

profekta te ekbar
kore importer dhore
then oita baad diye
babugula use kore

attribute closure ber
korte kore. Jodi dekh oita
chara generate hoeche, taile
baad.

$C \rightarrow A$
 $A \rightarrow B$
 $D \rightarrow E$
 $AC \rightarrow D$

$\therefore F_c = \{C \rightarrow A, A \rightarrow B, D \rightarrow E, AC \rightarrow D\}$

$(C, A, B, D, E) = 2$

$(C \rightarrow A, A \rightarrow B) = 2$

$(101, 102, 103) = 2$

Normalization Form :

1) 1NF = (16)

2) 2NF = (18)

3) 3NF $\rightarrow$ Lossless Decomposition, Dependency Preserve - 100%

4) 3.5 NF / Boyce-Codd NF $\rightarrow$ LD, DP - 50%

BCNF

5) 4NF

Employee			ID name		name salary	
ID	name	salary				
101	A	100K	101	A	A	100K
102	B	150K	102	B	B	150K
103	A	200K	103	A	A	200K

$R(A, B, C, D, E), F = \{A \rightarrow B, C \rightarrow E\}$

$R_1 \rightarrow F_1 = \{A \rightarrow B\}$

$R_2 \rightarrow F_2 = \{C \rightarrow E\}$

if both are

present in the decomposed tables, then dependency is preserved.

no. of employees having like thus

info loss korese. Emn habena LO er khetre

$R = \{A, B, C, D, E\} \rightarrow$ shob single elements oia math go

$R = (A, B, C, \dots) \rightarrow$ thake 1NF e nai

* BCNF:

$$R = (A, B, C, D, E)$$

$$F = \{A \rightarrow B, BC \rightarrow D\}$$

$$\begin{aligned} A &\rightarrow B \\ B &\rightarrow C \\ C &\rightarrow D \\ A &\rightarrow DA \end{aligned}$$

✓ Step 1 (Canonical cover)

Step 1:

$$A \rightarrow B$$

$$BC \rightarrow D$$

Step 2:

$$A \rightarrow B$$

$$BC \rightarrow D$$

Step 3:

$$A \rightarrow B$$

$$BC \rightarrow D$$

↑

$$B? (C^+) = C \times$$

$$C? (B^+) = B \times$$

Step 4:

$$A \rightarrow B$$

$$BC \rightarrow D$$

$$(A^+) = A, B \times$$

$$(BC)^+ = BC \times$$

Step-2

$$R = (A, B, C, D, E)$$

$$F_c = \{A \rightarrow B, BC \rightarrow D\}$$

So, Now we have to check whether the given R, F_c is in BCNF form or not. For this we have to find the attribute closure of the determinants. If all of them are super key, it is in BCNF form.

$(A)^+ = AB$, it is not a super key, so it is not in BCNF form.

$\boxed{A \rightarrow B}$ $(A, B, C, D, E) = \{A, B, C, D, E\}$

Step 3:

$R = (A, B, C, D, E)$

Always data table

$\left\{ \begin{array}{l} \rightarrow R_1 = (A, B) \text{ (jigular janna BCNF form break karna)} \\ \rightarrow R_2 = (C, D, E, A) \text{ (bakigula) } \rightarrow \text{foreign key or determinant} \end{array} \right.$

ALWAYS!

$F_1 = \{A \rightarrow B\}$

$A \rightarrow B$

$\therefore AC \rightarrow D$

[pseudo transitivity axiom]

$F_2 = \{AC \rightarrow D\}$

$AC \rightarrow D$

$\therefore$ dependency preserved.

for R_1 ,

$(A)^+ = AB$; it's SK hence in BCNF

for R_2 ,

$(AC)^+ = ACD$; it's not SK, hence violates the BCNF

$\therefore R_2 = (C, D, E, A), F_2 = \{AC \rightarrow D\}$

$\rightarrow R_3 = (A, C, D)$ $F_3 = \{AC \rightarrow D\}; (AC)^+ = ACD$; SK and in BCNF.

$\rightarrow R_4 = \text{~~(E, A, C)~~ } (E, \overbrace{A, C}^{\text{PK}}), F_4 = \{ \}$; by default BCNF
empty set

BCNF: $R_1 = (A, B), F_1 = \{A \rightarrow B\}$

$R_3 = (A, C, D), F_3 = \{AC \rightarrow D\}$

$R_4 = (E, A, C), F_4 = \{ \}$

$\therefore 3 + 2 = 5$

$R = (A, B, C, D, E)$

$R_1 = (A, B)$

$R_3 = (A, C, D)$

$A \rightarrow B$

$AC \rightarrow D$

$F_1 = \{A \rightarrow B\}$

$F_3 = \{AC \rightarrow D\}$

for R_1

$(A) = AB$; it is 2K hence in BCNF

for R_3

$(AC) = ACD$; it is not 2K hence not in BCNF

$(ACD) = ACD$; it is not 2K hence not in BCNF